THOMAS M. FRANCIS

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Portfolio: https://tmfrancis.github.io/

SUMMARY

Geospatial Data Scientist with demonstrated strengths in:

- data acquisition, validation, conversion, application, and analysis
- iintegrating raster, vector, and non-spatial data
- spatial data visualization

Proven communicator with excellent analytical skills. Enthusiastic and effective team player.

SKILLS

GIS Software: ArcGIS Pro, ArcGIS Online, Google Earth Pro, Survey123, QGIS, GeoServer, Leaflet
Programming Languages: R, Python, Arcade, SQL
Software: Revit, GitHub, MS-Excel, MS-Powerpoint, Zotero

EXPERIENCE

FACILITIES INFORMATION SPECIALIST, University of Massachusetts, Lowell, MA — 2026-Present

Manage spacial facilities information, and ESRI Enterprise and ArcGIS Online portals. Develop data collection policy and applications. Visualize planning and facilities management data to support institutional decision making.

GIS SPECIALIST & SUBJECT MATTER EXPERT, Industrial Economics Inc, Cambridge, MA — 2022 – 2025

Managed geospatial analyses and data sets for multiple environmental damage practices in an environmental consultancy. Developed science-based analyses and presentation materials to support Federal agency decisions, environmental damage assessments, and environmental impact statements. Trained and supported staff in geospatial software and analysis techniques.

RESEARCH ASSISTANT, Clark University, Worcester MA — 2022

Supported development of quantitative analysis metrics for Brazilian land use dataset. Researched metrics of spatial data set accuracy assessment for academic publication.

TEACHING ASSISTANT, Computer Programming for GIS — 2021

TEACHING ASSISTANT, Python Programming — 2021

Led lab sessions for 10-15 students and held office hours to assist with coursework. Graded lab assignments for undergraduate, master's, and doctoral students.

SATELLITE IMAGERY AND GEOGRAPHIC INFORMATION SYSTEMS CONTRACTOR, United Nations Institute for Training and Research, UNOSAT, remote — 2021

Investigated and documented petroleum capacity, production, and transfer in Central Africa using remote sensing. Delivered detailed, actionable analysis based on a diverse set of geospatial and non-spatial datasets.

DEPUTY DIRECTOR, Massachusetts Bicycle Coalition, Inc, Boston MA — 2016 – 2020

Managed essential programs for small non-profit with statewide mandate. Coordinated with federal, state, and local government officials and advocacy groups to improve transportation infrastructure. Active member of MassDOT Design Review Committee and Massachusetts Bicycle and Pedestrian Advisory Board.

OIL AND GAS ANALYST, Fossil Free Indexes, LLC, Port Washington NY — 2014 – 2017

Developed algorithms for identifying and ranking fossil fuel reserves held by public corporations. Maintained database of industry specific operational statistics. Interpreted scientific literature and provided timely written commentary on fossil fuels and climate change.

EDUCATION

MASTER OF SCIENCE, GEOGRAPHIC INFORMATION SCIENCE
Clark University, Worcester, MA

STRATEGIC MARKETING EXECUTIVE PROGRAM
Cornell University, Johnson School of Management, Ithaca NY

BACHELOR OF ART, ECONOMICS
Swarthmore College, Swarthmore, PA

PUBLICATION

Pontius, R. G., Jr, Francis, T., & Millones, M. (2025). A call to interpret disagreement components during classification assessment. *International Journal of Geographical Information Science*, 1–18. https://doi.org/10.1080/13658816.2025.2469830